

Simulation of Optical Transmission, Reflection and Absorption Spectra of ZnO Thin Films on Glass Using OpenFilters

Software: OpenFilters 1.1.1 (Free)

Duration: 2 Hours

Objectives

- Understand the optical properties of thin-film materials.
- Simulate transmission and reflection spectra using OpenFilters.
- Calculate absorptance from simulated spectra.
- Investigate the influence of thin-film thickness on optical properties.

Learning Outcomes

- Explain the optical behaviour of thin films.
- Build multilayer structures in OpenFilters.
- Simulate transmission and reflection spectra.
- Calculate absorptance.
- Analyse the effect of thickness on optical performance.

Software Required

OpenFilters Version 1.1.1

Materials Used

- Void
- Glass (BK7)
- ZnO

Thin Film Structure

Void

↓

ZnO (Variable Thickness)

↓

Glass (BK7)

Film Thicknesses

50 nm

100 nm

200 nm

400 nm

Simulation Parameters	Value
Wavelength Range	300–900 nm
Angle of Incidence	0°
Polarization	Unpolarized
Substrate	Glass

Procedure

A. Create a New Project

Step 1 — Start OpenFilters

You should see the main window with the menu bar:

File | Project | Filter | Analyse | Design/Optimize | Preproduction | Materials | Help

Step 2 — Create a new project

From the top menu, click:

File → New Project

A new project will be created.

B. Add a Filter

Step 3 — Add a filter to the project

From the top menu, select:

Project → Add Filter

A **Filter Properties** window/dialog box should appear.

This is the important window where you define the optical system.

C. Define the Filter Properties

Your ZnO sample is:

Void → ZnO → BK7 (Glass)

In the **Filter Properties** window, set the following parameters.

Step 4 — Select the substrate

Find the **Substrate** option.

Select:

BK7 (Glass)

The substrate is the material underneath the ZnO film.

Step 5 — Select the incident/front medium

Find the **Front medium** or **Medium** option.

Select:

void

Therefore, light will enter the sample from:

void → ZnO → BK7

OpenFilters' filter-properties dialog allows the substrate and front/back media to be specified at this stage.

Step 6 — Set the wavelength range

In the wavelength settings, enter:

From wavelength = 300 nm

To wavelength = 900 nm

Set the wavelength increment/resolution to:

1 nm

Thus, the simulation will calculate the optical properties from:

300 nm to 900 nm

Step 7 — Set the reference wavelength

OpenFilters also has a **reference/center wavelength**.

For this experiment, use:

Reference wavelength = 550 nm

This is a reference wavelength used by OpenFilters for displaying index profiles and optical thickness; it is not the wavelength range of the simulation.

Step 8 — Set the angle of incidence

Set:

Angle of incidence = 0°

This represents normal incidence.

Step 9 — Set polarization

For this experiment, use:

Unpolarized light

Step 10 — Backside consideration

Because your sample is:

void → ZnO → BK7 glass

keeping the default setting

Step 11 — Finish filter properties

Check that the important settings are:

Setting	Value
Front medium	void
Substrate	Glass (BK7)
Wavelength range	300–900 nm
Wavelength increment	1 nm
Reference wavelength	550 nm
Angle of incidence	0°
Polarization	Unpolarized

Then click:

OK

D. Add the ZnO Layer

After clicking **OK**, you should return to the main OpenFilters window.

Your filter currently has the substrate and incident medium, but **no ZnO coating layer yet**.

Step 12 — Add ZnO_Query

From the top menu, select:

Filter → Add layer

A new **Add Layer** dialog should appear.

Step 13 — Select ZnO_Query material

In the **Material** field, select:

ZnO_Query

If ZnO_Query is available in the OpenFilters material database, select it directly.

Step 14 — Enter ZnO thickness

In the **Thickness** field, enter:

100 nm

Then click: **OK**

E. Check the Layer Structure

Now look at the upper table in the main window.

You should have:

Front medium: void

Front layer: ZnO, 100 nm

Substrate: BK7 Glass

After **Filter → Add layer → ZnO → 100 nm**, the number of front layers should become:

Nb front layers = 1

F. Calculate Transmission and Reflection

Now you are ready to perform the optical calculation.

Step 15 — Calculate transmission

From the menu bar, select:

Analyse → Calculate Transmission

OpenFilters provides calculation of transmission, reflection and absorption as part of its analysis capabilities.

The graph at the bottom of the OpenFilters window should now display the calculated transmission spectrum.

Step 16 — Calculate reflection

Next select:

Analyse → Calculate Reflection

The reflection spectrum will be calculated.

The official getting-started guide uses the same **Analyse → Calculate Reflection** workflow for calculating optical properties.

G. View the Results in the Bottom Graph

For this experiment, you need:

Photometric

Click:

Photometric

This is the relevant section for viewing the optical transmission/reflection results.

H. Calculate Absorption

You can calculate absorption using:

$$A = 1 - T - R$$

where:

- A = absorptance
- T = transmittance
- R = reflectance

OpenFilters itself is capable of calculating absorption in addition to transmission and reflection.

I. Save the 100 nm Results

Step 17 — Save the project

Use:

File → Save Project

Save it as:

ZnO_100nm.ofp

Step 18 — Export the numerical results

Analyse → Export results as text

Keep these files because they will be used to prepare the observation table and compare the different thicknesses.

J. Repeat for Other ZnO Thicknesses

Now you **do not need to create a new project every time.**

Simply change the ZnO layer thickness and recalculate.

Repeat for-

50 nm

200 nm

400 nm

Observation Table

Thickness(nm)	Maximum T(%)	Minimum T(%)	Maximum R(%)
50			
100			
200			
400			

Discussion Questions

- Why does ZnO exhibit high transparency?
- How does film thickness influence transmission?
- Is this material suitable as a window layer? Explain.
- Why is glass used as the substrate?
- Suggest two applications of ZnO.